Mumford congruence under affine normalization 2026-10-31

T-1135. Producer formal proof packet; independent source-fidelity review pending.

Construction

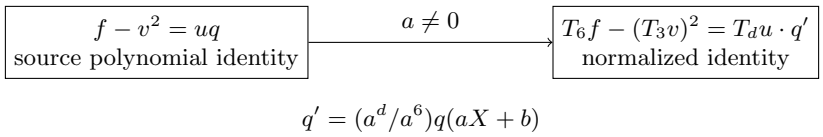
Over a field K, set $T_n(p) = a^{-n}p(aX + b)$. For $a \neq 0$ and any natural d, the checked transport is

$$u \mid f - v^2 \quad \Longrightarrow \quad T_d(u) \mid T_6(f) - T_3(v)^2.$$

When $f - v^2 = uq$, the explicit quotient on the right is

$$\frac{a^d}{a^6}\, q(aX + b).$$

Separate declarations check square/difference compatibility, evaluation, root preservation, identity support and zero ordinate.



$$q' = (a^d/a^6)q(aX + b)$$

Fixed controls

Use $f = X^5 + 1, u = X, v = 1, q = X^4, a = 2, b = 3, d = 1$. At the transformed support root $-3/2$, the ordinate is $1/8$ and the curve value is $1/64$. The correct residual is zero. Omitting ordinate scaling yields $-63/64 \neq 0$. The zero-scale control shows coordinate collapse; that scale is excluded by the main theorem. A fixed divisibility instance is also checked.

Source connection and remaining bindings

The historical `genus2_model_gate.rs`, lines14–56, uses these substitution/scaling factors. The preceding Scholten model packet supplies the concrete affine equation. Literal Rust-loop/compiler correspondence, monicity and degree bounds, projective transport and divisor-class interpretation are separate next bindings.

Evidence

Nine main declarations and nine controls have separate named axiom readback. Raw attempts, native exit codes, unchanged dependency/source bindings and standardized status accompany this report. Attempt1 failed on polynomial syntax and a looping simplifier; attempt2 corrected them. The first control compile timed out after120 seconds; the identical retry passed. Nonfatal warnings are retained. Existing compiled Lean dependencies remain part of the trust boundary.

Primary documentation: Mathlib polynomial evaluation/composition. No novelty claim. No scientific execution, timing, enumeration or cost ratio; expired campaign limits unchanged. The diagram describes algebraic dependencies, not measured performance.